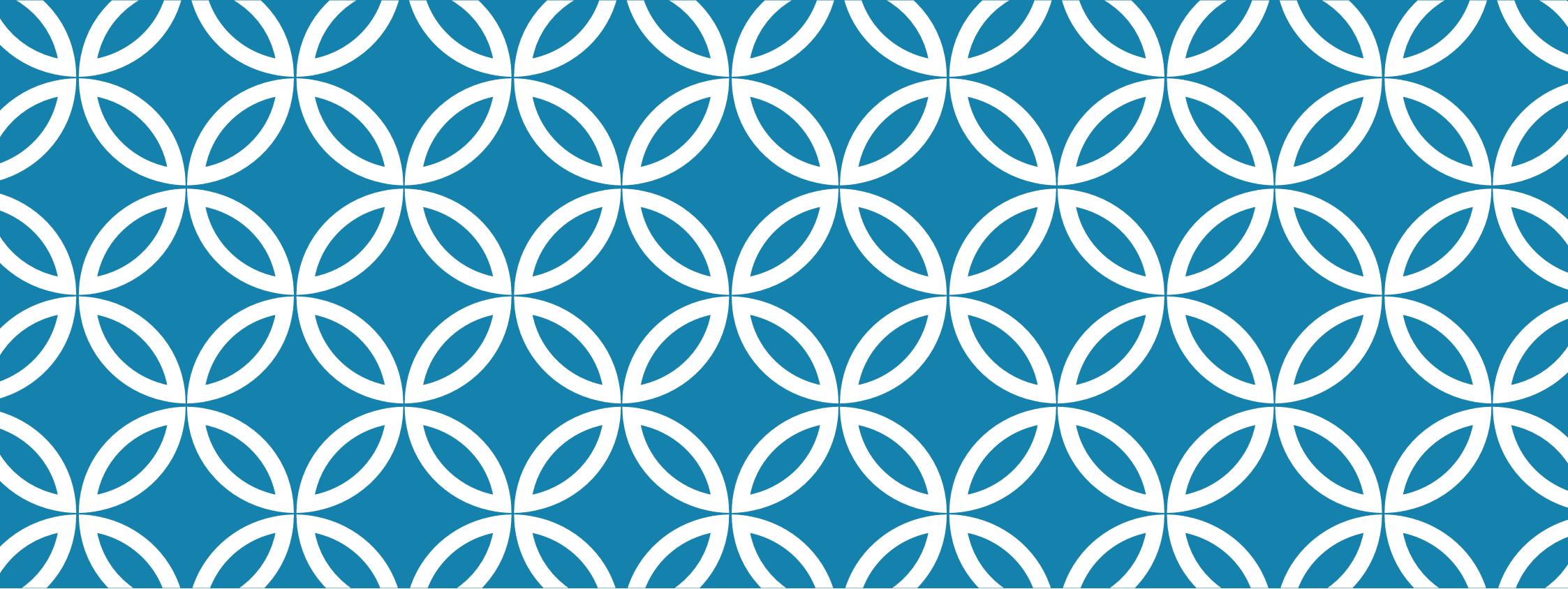




ESTIMATION OF A LANDSCAPE PROJECT

Lecture VII



Edit with WPS Office

Landscape Planting and Irrigation play an important role

Landscaping provides cost savings like site finishes, natural drainage, storm water pollution prevention, solar heat reduction etc.

Cost estimate will depend upon:

- i. Quantities
- ii. Labour
- iii. Material
- iv. Equipment
- v. Indirect costs

Influencing factors are:

- i. Who is the client/ owner
- ii. Location, climate
- iii. Scale of the project
- iv. Availability of labour and materials
- v. Taxes, wages, rates etc.

Green building and sustainable projects will have other design criteria.

- ✓ They may reduce irrigation water usage for planting and use reclaimed or grey water for irrigation.
- ✓ To do the estimation, the estimator should take it in the sequence of works on site from beginning to end.
- ✓ The estimator may also use grouping methods like irrigation works + planting works.



Edit with WPS Office

Quantity Control

For double checking of figures,

If overall site area is 107,000 sq ft, and the landscaping area is 35,000 sq ft., it can be taken as 33%.

Cross-checking the accuracy therefore, becomes easy. For example, 20,000 sq ft. of hydroseed and 15,000 sq ft. of sod is used, so total 35,000 is almost equal to the control quantity of the planted area.

Landscape sub-contractor does:

- ✓ **Top soil** placement and soil preparation – Top soil is either imported from off site or recovered from excavation by earthwork contractor. Unit used is cubic yards or cubic metres. Coverage area (in sq ft.) is multiplied by depth.
- ✓ **Irrigation** pipe length is measured in running metres.
- ✓ **Economics of scale** is also applicable so larger projects will be more cost efficient than smaller projects, as labourers have less opportunity to create an efficient work flow.
- ✓ **Location** affect wage of labours, taxes, freight charges, and indirect costs. Rainfall will also vary so it will impact the irrigation plan.

Application of LEED Principals

- ✓ **Recycled water** will be used, smart irrigation controllers such as, **weather controllers** can be used. **Thermostats** control the water supply as opposed to the standard controllers which works on a particular schedule.
- ✓ A typical **landscape crew** includes labourers, skilled workers and light equipment operators; different works have to be analysed to see **number and type of labourers** needed and **number of hours** required for the work.
- ✓ Materials and plants used – If non-natural plants are used instead of natural plants, cost will vary.
- ✓ **Equipments** are accounted for on **hourly rate**, so equipment renting cost + fuel cost + maintenance cost.
- ✓ **Type of theme** – Selection of trees, perennials, annuals etc. Climate zone, soil typology, plants growing patterns are considered.
- ✓ **Type of grass selection** depends on temperature, rainfall, foot traffic, soil, proximity to river or coastal areas.

Buffalo grass, zoysia, bent, creeping bent grass, sheep's fescue, creeping red fescue, tall breed fescue, red fescue, emerald zoysia etc.

3 components of soil type: sand, silt and clay.

- ✓ Sand – 0.05 to 2 mm, drainage properties due to pore space, water percolates through.
- ✓ Silt – 0.05 to 0.002 mm, holds water, texture feels smooth.
- ✓ Clay - <0.002 mm, water holding capacity is high; too much clay is difficult to manage, water can be retained for many days.

Type of irrigation required will depend on type of soil.

Sprinkler system can be of two types:

Pop-ups:

- ✓ Pop-ups are more popular, they turn on and retract back down when no water is running through.
- ✓ Looks wise, it is appealing to the eye and very functional.
- ✓ User-friendly as owners can easily mow the lawn without any damage.

Rotor sprinklers used for sod area:

- ✓ These have larger coverage radius ranging from 15 to 35 ft., compared to pop-ups (8 to 15 ft.)
- ✓ So, rotor sprinklers will require half amount of heads, whereas pop-ups will require more.

- ✓ For smaller sod areas, pop-up sprinklers are better.
- ✓ Main lines, valves and lateral lines are also vital to an irrigation system.
- ✓ Valves can help compare which area of the lawn needs to be irrigated.

Plans required for

- Irrigation.
- Planting.
- Detailed section drawings of pathways.
- Landscape plan.
- Grading Plan.
- Drainage Plan.
- Illumination Plan.

List of works:

- Earthwork.
- Importation of top soil.
- Finish grading landscaping area.

Trenches for irrigation:

- Excavation of trenches.
- Bankfill and compaction of trenches.

For example, an irrigation plan consists of valves, bank-flow prevention, pop-up sprinklers, rain sensors and trenches.

Irrigation:

- Pipe sleeves
- Valves, sheet of valves
- Control wing
- Sprinklers
- Water metres etc.
- Cost of performing check tests.

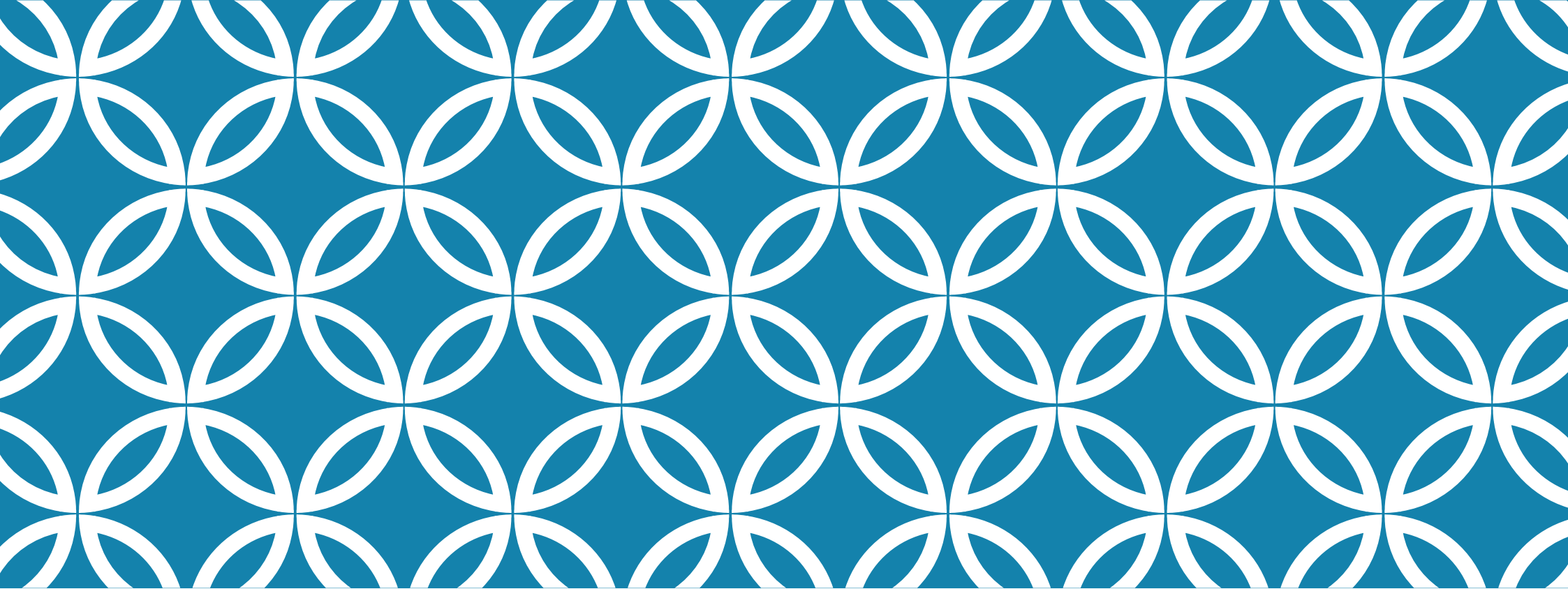
Landscape planting:

- Trees (all species), flower plants, shrubs, vines.
- Sod.
- Hydroseed.
- Other miscellaneous items (root barrier, bank mulch, fertilizers, brick on edge)
- Fences, pathways, drains etc

Estimating

To the ordinary person, estimation seems like a very simple thing. However, it entails multiple tasks that need to be incorporated in an estimate. Labour, materials, equipment, subcontracting, contingencies, sales tax, profit are all things that need to be incorporated to ensure a correct bid with a profitable outcome, which are described in full as follows:

- ✓ **Labour costs:** wages and benefits of work crew.
- ✓ **Material costs:** Materials required for the project. Plants, boulders, fertilizers, mulch, PVC, sprinklers, valves, controllers, wood, concrete etc.
- ✓ **Equipment costs:** Equipment needed to complete a project, whether it is rented , making payments, or completely owned. For example, tiller, backhoe etc.
- ✓ **Subcontracting costs:** Even if the landscape contractor is not doing a specific task onsite, it still needs to be included in the bid. Usually subcontracting occurs, if it falls out of the realm of contractor's expertise.
- ✓ **Contingencies:** This is a backup plan in case something comes up during a project that was not accounted originally for; this can range from missing certain tasks to missed items. However, contingencies account for only a fraction of the original bid, so if too many items are being forgotten, a landscape contractor loses a lot of money.
- ✓ **Profits:** Profits need to be included in a bid or the job is practically being done



THANK
YOU



Edit with WPS Office